

Introduction

AI Phishing Detection Platform

Simplified Web-Based Project Documentation

(Beginner-Friendly + Resume-Focused Version)

1. Introduction

The **AI Phishing Detection Platform** is a web-based cybersecurity application that helps users identify whether a URL, email, or message is safe or potentially phishing-related.

Phishing is one of the most common cyberattacks where attackers create fake websites, emails, or messages to steal sensitive information such as:

- Passwords
- Banking details
- OTPs
- Personal information

This project focuses on creating a practical and beginner-friendly phishing detection platform using modern web development technologies and external security APIs instead of building complex AI models from scratch.

The system analyzes:

- URLs
- Email content
- Suspicious messages

and provides:

- Risk scores
- Safety warnings
- Threat reports
- Detection history

This project is ideal for a fresher because it demonstrates:

- Full Stack Development

- API Integration
- Cybersecurity concepts
- Dashboard development
- Real-world problem solving

without requiring advanced Machine Learning knowledge.

2. Problem Statement

Many internet users cannot easily identify fake or malicious websites and emails.

Cybercriminals create phishing attacks that look genuine and trick users into:

- Sharing passwords
- Entering banking details
- Downloading malware

The goal of this project is to create a web application that helps users:

- Verify suspicious links
 - Analyze emails
 - Detect possible phishing attempts
 - Improve cybersecurity awareness
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3. Objectives of the Project

The main objectives are:

- Build a phishing detection web application
 - Analyze suspicious URLs
 - Analyze email/message content
 - Generate risk analysis reports
 - Store scan history
 - Create a user dashboard
 - Improve cybersecurity awareness
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4. Detailed Explanation of the Project

4.1 URL Scanner

Users can paste a website URL into the application.

The system checks:

- HTTPS usage
- Suspicious keywords
- Domain structure
- Blacklisted URLs
- URL reputation using security APIs

The platform then generates:

- Risk Level
- Threat Status
- Safety Recommendation

Example

Input:

`http://secure-paypal-login-free.com`

Output:

Risk Level: HIGH

Status: Suspicious Website

Recommendation: Avoid Visiting

4.2 Email Content Analysis

Users can paste email content into the platform.

The system checks for:

- Urgent/threatening language

- Suspicious links
- Fake security warnings
- Credential requests
- Scam-related keywords

Example

Input:

“Your bank account will be suspended. Verify immediately.”

Output:

Risk Level: HIGH

Possible Phishing Attempt Detected

4.3 Risk Analysis System

Instead of training custom AI models, the project uses:

- Rule-based detection
- Security APIs
- Keyword analysis
- URL reputation services

This approach is:

- Easier for beginners
- Faster to build
- More practical for a fresher-level project

The platform calculates:

- Risk percentage
 - Suspicion level
 - Warning messages
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4.4 Dashboard

Users can access a dashboard showing:

- Total scans
- Safe URLs
- Suspicious URLs
- Scan history
- Detection statistics

The dashboard improves:

- User experience
 - Data tracking
 - Threat visibility
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5. Core Features

User Features

- User Authentication
- URL Scanner
- Email Analyzer
- Risk Detection
- Scan History
- Dashboard
- Threat Reports

Admin Features

- User Management
 - View Detection Logs
 - Monitor Reports
 - Manage Suspicious Keywords
 - View Analytics
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6. User Roles Explanation

6.1 User

A normal user can:

- Register/Login
 - Scan URLs
 - Analyze email content
 - View reports
 - Access scan history
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6.2 Admin

The admin manages the system.

Admin capabilities:

- Monitor user activity
 - View analytics
 - Manage reports
 - Update phishing keyword lists
 - Track suspicious activity
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7. Use Cases

Use Case 1 — URL Verification

A user receives an unknown website link and checks whether it is safe before visiting.

Use Case 2 — Email Safety Check

A user copies suspicious email content into the system for analysis.

Use Case 3 — Cybersecurity Awareness

Students use the platform to learn about phishing attacks and online safety.

Use Case 4 — Organization Usage

Companies can use the platform internally to verify suspicious links shared by employees.

8. Importance of the Project

This project is important because:

- Phishing attacks are increasing rapidly
- Many users lack cybersecurity awareness
- Businesses need safer browsing practices

The project helps:

- Improve online safety
 - Prevent scams
 - Protect user data
 - Increase cybersecurity awareness
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Project designing Info

9. Technologies Used

Frontend

- Next.js
- React.js
- Tailwind CSS
- Framer Motion

Backend

- Node.js + Express.js

Database

- PostgreSQL
or
- Supabase

APIs & Services

- Google Safe Browsing API
- VirusTotal API

Authentication

- JWT Authentication
or
 - Clerk Authentication
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10. System Architecture

User → Frontend → Backend API → Security APIs → Risk Analysis → Dashboard

11. Database Design

Users Table

- id
- name
- email
- password
- role

Scan History Table

- id
- user_id
- scan_type
- content
- result
- risk_level
- created_at

Reports Table

- id
 - report_type
 - details
 - created_at
-

12. Project Workflow

Step 1

User submits URL or email content.

Step 2

Backend receives request.

Step 3

System checks:

- Keywords
- URL structure
- Security APIs

Step 4

Risk score is generated.

Step 5

Results are displayed on dashboard.

Information for you

13. Impact on Resume

This project can significantly improve a fresher's resume because it demonstrates:

Technical Skills

- Full Stack Development
- REST APIs
- Authentication
- Database Management
- Dashboard Development
- API Integration

Cybersecurity Knowledge

- Phishing detection concepts
- Security analysis
- URL verification
- Threat awareness

Real-World Problem Solving

This project solves an actual cybersecurity problem faced by users and organizations.

14. Why This Project is Good for Freshers

This version is beginner-friendly because:

- No advanced AI training is required
 - No complex machine learning setup
 - Focuses on practical development skills
 - Easier to complete within 30 days
 - Strong enough for interviews and portfolios
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15. Future Enhancements

Future improvements can include:

- Browser extension
 - AI-based prediction models
 - Real-time email integration
 - Mobile application
 - Multi-language support
 - Live threat monitoring
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Schedule

16. 30-Day Development Schedule

Week 1 — Planning & Setup

Day 1

- Finalize project requirements
- Research phishing detection systems

Day 2

- Design database schema
- Create project architecture

Day 3

- Setup Next.js frontend

Day 4

- Setup Node.js backend

Day 5

- Configure database

Day 6

- Setup authentication

Day 7

- Create homepage UI
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Week 2 — Core Development

Day 8

- Build login/register system

Day 9

- Create dashboard UI

Day 10

- Build URL scanning feature

Day 11

- Integrate VirusTotal API

Day 12

- Integrate Google Safe Browsing API

Day 13

- Display risk analysis results

Day 14

- Store scan history
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Week 3 — Email Analysis & Admin Panel

Day 15

- Create email analysis feature

Day 16

- Add keyword detection logic

Day 17

- Create reports section

Day 18

- Build admin dashboard

Day 19

- Add analytics cards

Day 20

- Add filters/search

Day 21

- Testing and debugging
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Week 4 — Finalization & Deployment

Day 22

- Improve UI/UX

Day 23

- Add loading states and alerts

Day 24

- Add responsive design

Day 25

- Improve validation/security

Day 26

- Optimize APIs

Day 27

- Deploy frontend

Day 28

- Deploy backend

Day 29

- Final testing

Day 30

- Documentation
 - Resume preparation
 - GitHub optimization
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Resources

17. Resources List

Frontend Resources

Next.js

[Next.js Documentation](#)

Tailwind CSS

[Tailwind CSS Documentation](#)

React

[React Documentation](#)

Backend Resources

Node.js

[Node.js Documentation](#)

Express.js

[Express.js Documentation](#)

Database Resources

PostgreSQL

[PostgreSQL Documentation](#)

Supabase

[Supabase Official Website](#)

API Resources

VirusTotal API

[VirusTotal API Documentation](#)

Google Safe Browsing API

[Google Safe Browsing API Documentation](#)

Cybersecurity Resources

OWASP

[OWASP Official Website](#)

Learning Resources

FreeCodeCamp

[freeCodeCamp YouTube Channel](#)

Traversy Media

[Traversy Media YouTube Channel](#)

CodeWithHarry

[CodeWithHarry YouTube Channel](#)

Conclusion

18. Conclusion

The AI Phishing Detection Platform is a practical and modern cybersecurity web application designed for beginner and intermediate developers.

The project helps users:

- Detect suspicious websites
- Analyze phishing emails
- Improve online safety
- Learn cybersecurity concepts

This simplified version focuses on:

- Full Stack Web Development
- API Integration
- Dashboard Development
- Real-world cybersecurity problem solving

without requiring advanced AI or Machine Learning knowledge.

For a fresher, this project is highly valuable because it demonstrates:

- Industry-relevant skills
- Practical application development
- Security awareness
- Professional project architecture

This project can strongly improve a resume and portfolio while preparing developers for real-world software development roles.